

Two-Stage Water Rocket & Its Simulation Platform

Transcultural & transectoral collaboration in transdisciplinary education

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Significance — Why it matters

How do you teach abstract mechanics to young children across a language and sector boundary? We designed a **bilingual (Chinese–English)** STEM unit and delivered it as a **cross-regional service-learning project** — Hong Kong university students teaching primary-school children in mainland China. The water rocket becomes a safe, low-cost “living laboratory” for force, pressure and acceleration.

Transdisciplinary physics × engineering × AI × pedagogy **Transectoral** university ↔ school ↔ community **Transcultural** Hong Kong ↔ Jiangxi, bilingual



Objectives

Knowledge

Explain launch principles, pressure vs. air pressure, action–reaction, acceleration and air resistance.

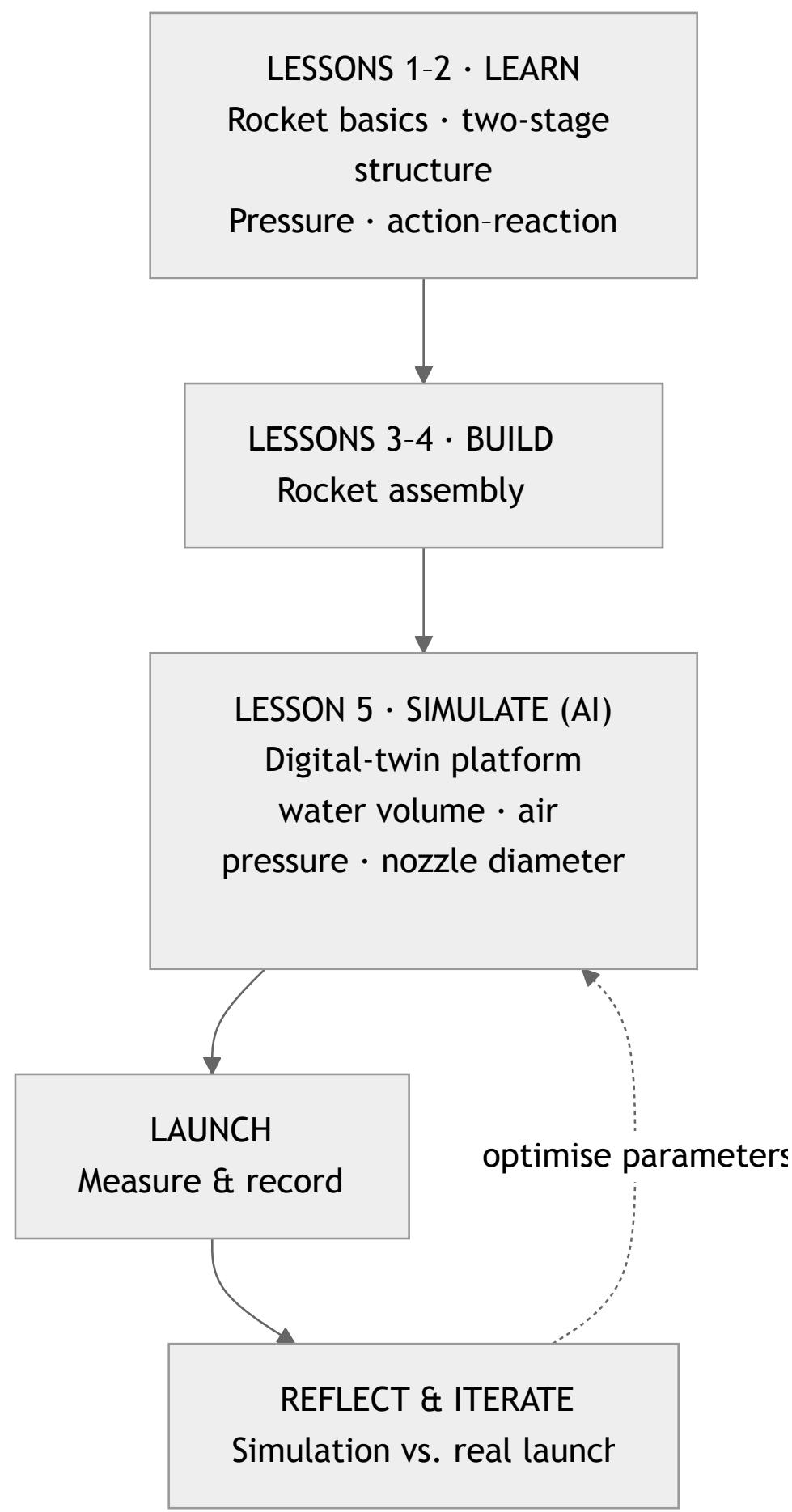
Ability

Practise **hypothesis** → **test** → **optimise**; variable control; data recording; engineering design.

Transdisciplinary

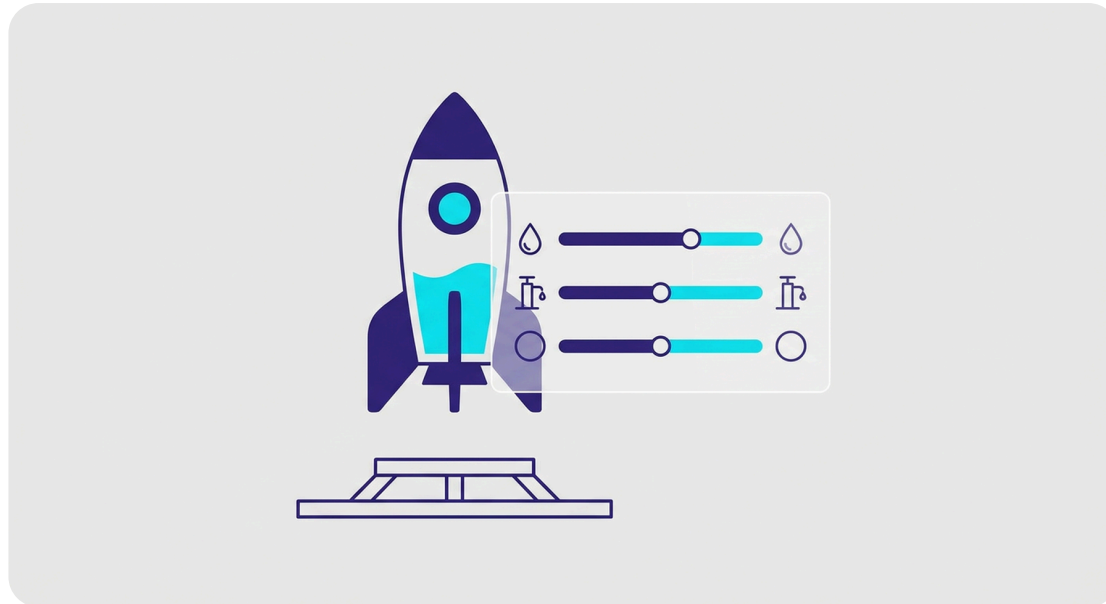
Model a pedagogy coupling **AI simulation** with **hands-on experiment**, transferable to other STEM topics.

Approach & Methodology



A custom **AI simulation platform** acts as a **pre-launch “digital twin”**: students sweep **water volume**, **air pressure** and **nozzle diameter** to predict the optimal launch *before* the physical test — reducing cost and safety risk while grounding intuition in theory.

Delivered bilingually across five scaffolded lessons, co-taught by university students for primary-school learners.



Repurposed materials



Building together



Hands-on assembly

Key Findings & Implications

Anticipated outcomes By the end of the unit, a target of **[X]**% of learners are expected to explain action–reaction and pressure; simulation predictions will be validated against **[M]** real launches.

Stakeholder impact **Learners** grasp core mechanics and build inquiry skills; the **school & community** gain a reusable, low-cost STEM module; we, as **facilitators**, develop transdisciplinary teamwork and cross-cultural communication.

Next steps Map physics one-to-one onto build/launch steps; add academic-standard scenarios; build a student innovation zone to close the loop **learn** → **practise** → **create**.



“The children connected pressure and force faster than any worksheet ever did.”

— Teacher testimony (to be confirmed)

“I learned why my rocket flew higher when I pumped in more air.”

— Student testimony (to be confirmed)

References & Acknowledgement

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Learn more / full materials: github.com/tesolchina/paste2026-poster — lesson plan, AI simulation platform and photo gallery.



Scan to explore the full project

github.com/tesolchina/paste2026-poster

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